

NIRF ML Analytics & Prediction Platform (Detailed Report)

Abstract

This project builds a web-based analytics and prediction platform that extracts NIRF ranking data from PDF reports, converts it into structured datasets, performs exploratory data analysis (EDA), applies statistical testing, and uses machine learning to detect ranking bias, identify feature importance, and forecast future trends.

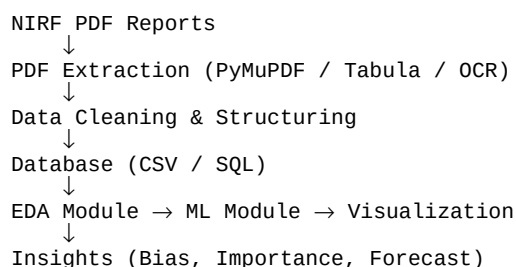
Problem Statement

- NIRF data is locked in unstructured PDF reports.
- No centralized structured dataset exists for analysis.
- Lack of transparency in how ranks are influenced.
- No system to detect bias or unfair patterns.
- No predictive system to forecast future rankings.

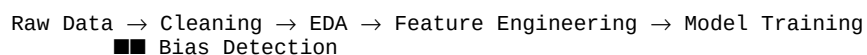
Objectives

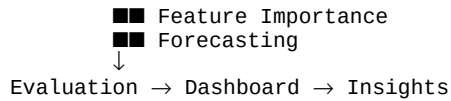
- Automate PDF to structured data extraction.
- Perform EDA to understand patterns and correlations.
- Detect bias using ML models.
- Identify most influential ranking parameters.
- Forecast future ranking trends.
- Build a web dashboard for visualization.

System Architecture



ML Pipeline





Exploratory Data Analysis (EDA)

- Distribution of scores (TLR, RPC, GO, OI, PR)
- Correlation heatmaps
- Year-wise trends
- Outlier detection

Machine Learning Models & Justification

1. Bias Detection – Random Forest

- Handles non-linear relationships
- Robust and stable predictions
- Compares actual vs predicted rank

2. Feature Importance – XGBoost

- High accuracy
- Provides feature importance scores
- Captures complex relationships

3. Forecasting – ARIMA & LSTM

- ARIMA: Good for simple time-series trends
- LSTM: Captures long-term complex patterns

4. Baseline – Linear Regression

- Simple and interpretable
- Used for comparison

Bias Detection Method

Train a model to predict rank using features. Compare predicted rank with actual rank. Large deviations may indicate bias in ranking.

Feature Importance

Use tree-based models to compute importance scores of each parameter to understand which factors influence ranking most.

Trend Forecasting

Use historical data (2016–2025) to predict future ranks and scores for institutions.

Evaluation Metrics

- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)
- R^2 Score

Tech Stack

- Python (Pandas, NumPy, Scikit-learn)
- XGBoost, TensorFlow/PyTorch
- Matplotlib, Seaborn
- React (Frontend)
- Flask/Django (Backend)
- MongoDB / SQL

Conclusion

This platform enables automated extraction, deep analysis, fairness evaluation, and prediction of NIRF rankings, providing insights that are not visible in raw PDF data.